Auditing Weather-Feature Reliance in Detrended Crop-Yield Models

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***Abstract*—Interpretable machine learning cannot support substantive claims about below-trend crop-yield events unless its fitted predictive model has adequate out-of-sample fidelity. We audit this condition in a U.S. crop–state–year panel with 1257 rows for Barley, Canola, Oats, and Wheat from 1990–2025. Trends and residual scales are fit on training years only; selection uses 2012–2015 validation, and 2016–2025 remains locked until final evaluation. The validation-selected ExtraTrees weather only model has final $R^2 = -0.014$, RMSE 0.669 t ha^{-1}, and paired ΔRMSE versus a zero-residual baseline of -0.005 with upper 95% bound 0.009. The selected model fails Gate A, and weather features also fail the primary incremental-value Gate B1. We therefore retain SHAP, LIME, permutation, ablation, and response-curve results only as descriptions of a fitted function, not evidence that weather caused observed events. The contribution is a reproducible stop-gate protocol with paired year-block uncertainty, tail checks, robustness checks, and an explicit warning that full-series detrending is retrospective only.**

***Index Terms*—crop yield, explainable machine learning, temporal validation, negative results, reproducibility**

I. Introduction

Machine-learning yield forecasts are increasingly built from weather, remote sensing, management, and crop-model variables. Studies report useful predictive systems across large and small panels, diverse model families, and hybrid formulations [1]–[7]. Yet a plausible feature attribution is not causal evidence, and agreement between explanation methods cannot compensate for a model that does not predict the future target reliably. This distinction is especially important for below-trend yield events, where a compelling explanation of an individual fitted prediction can be mistaken for an explanation of the observed loss.

Weather extremes and climate variability are materially associated with crop-yield variation at several spatial scales [8]–[16]. Those findings motivate careful predictive evaluation; they do not license an event-level attribution from this restricted panel. We ask a narrower question: does a leakage-safe residual model have enough future-period fidelity and stability to permit substantive weather-feature-reliance interpretation? The answer is evaluated before XAI is interpreted.

The paper makes three contributions. First, it uses train-only target construction and validation-then-locked-test selection. Second, it specifies a quantitative fidelity gate with same-task baselines, paired year-block intervals, tail metrics, and pre-specified robustness checks. Third, it contrasts a prospective target with a retrospective full-series detrending diagnostic. A gate failure is a negative methodological result: it stops causal or event-driver claims, rather than choosing the most attractive explanation.

II. Related Work

A. Yield Modeling and Detrending

Detrending separates a local historical yield trajectory from residual variation, but the result can depend on the detrending procedure and information available at the evaluation date [17], [18]. We therefore fit each crop–state trend and residual scale on training observations only. Ridge regression, Random Forest, and ExtraTrees provide deliberately distinct linear and tree-based model families [19]–[21]; the implementations use scikit-learn [22]. This is a finite pre-specified comparison, not a search over the locked test period.

B. Explanation Methods and Their Boundary

SHAP and LIME explain a model output under their respective approximation and reference choices [23], [24]. Permutation-style analyses, model reliance, accumulated local effects, and conditional importance likewise describe feature dependence or behavior of a fitted predictor [25]–[27]. They can be useful diagnostic artifacts. However, none turns failed prospective predictive fidelity into evidence about an observed yield anomaly. The gate below formalizes this boundary.

C. Evaluation and Reproducibility

Random splits can be inappropriate when temporal or hierarchical structure is present [28]. Leakage can create seemingly strong scientific machine-learning results while invalidating the intended forecast claim [29]. Reproducible ML also requires explicit code, data paths, settings, and numerical evidence [30]. Accordingly, every manuscript number is generated from an auditable run: configuration, prediction vectors, bootstrap draws, citation check, and final PDF are all versioned outputs.

III. Data and Protocol

A. Panel, Features, and Target

The unit is a crop–state–year row. The processed panel contains 1257 observations from 1990–2025 for Barley, Canola, Oats, and Wheat across 12 U.S. states. Yield in t ha^{-1} comes from USDA NASS Quick Stats [31]; daily maximum/minimum temperature, precipitation, and shortwave radiation come from NASA POWER [32]. The primary design has 35 full-season weather features. A separate sensitivity design adds early-, middle-, and late-season proxies without redefining the target or locked population.

For a test interval, each crop–state linear trend and residual scale is estimated using only its associated training rows. A

TABLE I
DATA-FLOW AND FINAL-TEST ELIGIBILITY RECONSTRUCTED FROM THE RAW INPUTS.

Stage	Rows	Rule
raw yield file	1291	source table supplied in repository
analysis year filter	1257	exclude 34 raw 1989 rows because analysis starts in 1990
processed model frame	1257	exact reconstruction from raw yields plus NASA POWER daily input
final test eligible	333	2016-2025 rows with at least three prior series observations

row is eligible only with at least three training observations. The validation period contains 140 eligible rows after 4 pre-specified exclusions. The locked test starts with 343 candidates, excludes 10, and evaluates 333 rows. Future yields cannot change a previously constructed evaluation target; this property has a unit test and row-level audit. The learned target is the raw train-only residual in t ha^{-1}; the standardized residual $z = r/\hat{\sigma}_{\text{train}}$ defines below-trend events only. This distinction prevents a pooled raw-error score from being misread as a standardized event-recovery metric. Min-history rules of 3, 5, 8, and 10 years are reported as sensitivity analyses because both eligibility and the fitted residual scale can change with available history. Three observations are the minimum that leaves one positive residual degree of freedom for estimating a training residual scale; this is not claimed to provide a stable series-specific variance estimate.

B. Selection and Baselines

Ridge $\alpha \in \{1, 10\}$ and Random Forest/ExtraTrees with 160 trees and $\text{min\_samples\_leaf} \in \{1, 2\}$ are evaluated on metadata-only, weather-only, and full representations. Stochastic fits use fixed seeds $\{7, 17, 27, 37, 47\}$; numeric imputation/scaling, categorical encoding, and model fitting occur within the training fold. Mean per-row seed-aggregated validation RMSE chooses the configuration, with lexicographic tie-breaking and no locked-test access. Seed-level RMSE standard deviations are reported so tiny differences are not presented as deterministic wins.

The final audit compares learned predictions on identical rows to zero residual, training-mean residual, and crop training-mean residual baselines. Prediction vectors are hashed and pairwise maximum absolute differences are retained, so any nearly coincident baselines are demonstrated rather than assumed. We use 2,000 deterministic calendar-year block bootstrap replicates and percentile 95% intervals. An invalid replicate fails the run; none are silently dropped. We do not implement the stationary bootstrap itself; we cite it for the dependent-resampling motivation and resample complete calendar years as paired blocks [33]. State-year and crop–state cluster schemes are reported as pre-specified sensitivity analyses.

C. Pre-specified Fidelity Gate

For a paired comparison, model-minus-baseline error passes only when the upper 95% year-block interval is below zero. Gate A is predictive fidelity of the validation-locked choice. Gate B1 is the primary incremental-weather-value test: the validation-selected Weather only versus Metadata only contrast. Gate B2 (Full versus Metadata only) is sensitivity only and never triggers re-selection. Weather-specific claims require both Gate A and Gate B1. The primary operational tail is $z < -1$; $z < -1.5$ and $z < -2$ are sensitivity analyses because their smaller samples reduce power. Sign accuracy on a negative-only subset is reported only as a diagnostic, not a gate component. Instead, rank recovery requires a positive lower bootstrap bound and a within-year permutation $p < .05$. Top-k recovery reports random expectation k/n, lift, hypergeometric probability, permutation p, and a year-block lift interval. Thus a severe-event row cannot pass simply by overlapping one arbitrary event. The fixed policy is in Table II.

Algorithm 1: Locked fidelity audit
Input: panel, fixed split/configuration, seed list, gate configuration.
1. Fit detrending and preprocessing on each training period only.
2. Select one configuration on validation RMSE; lock it.
3. Refit the locked choice through 2015 and predict 2016–2025.
4. Compare same-row baselines with paired year-block bootstrap intervals.
5. Evaluate null-aware primary-tail rank/top-k recovery and sensitivity tails.
6. Return Gate A, primary Gate B1, and sensitivity Gate B2 separately. Substantive observed-event claims require Gate A, while weather-specific claims require both Gate A and Gate B1. Gate B2 cannot trigger model re-selection or replace Gate B1.

IV. RESULTS

A. Validation and Locked Test

Table III shows leading validation configurations. The selected model is ExtraTrees with weather only features. The complete grid, seed-level metrics, tie rule, and locked-test access flag are supplied in the artifact. A non-selected ExtraTrees full model can have a small final point estimate without changing the decision: it was not validation-selected, and the locked test is never used to switch configurations.

The selected weather-only ExtraTrees configuration ranks first in 3/5 stochastic validation seeds. Its advantage is small relative to several seed standard deviations, so the selected row is a reproducible protocol choice rather than a claim of decisive superiority. The one-standard-error flag and every seed-level score are retained; no final-test statistic changes this selection.

On the locked test, the selected model has $R^2 = -0.014$ and RMSE 0.669 t ha^{-1}. Its paired RMSE difference from zero crosses zero, so Gate A fails. Table V includes all same-task naive baselines and the selected configuration across feature families. Figure 2 separates three locked-test questions. Gate A evaluates whether the validation-selected weather-only model improves on the zero-residual baseline. Gate B1, the primary incremental-weather test, compares the same selected weather-only model with metadata-only context. Gate B2 compares the full and metadata-only representations as a sensitivity analysis; it cannot trigger re-selection or replace Gate B1. All comparisons use paired predictions on identical locked rows. Gate A, Gate B1, and Gate B2 all fail.

B. Event-Detection Diagnostics

Event detection is evaluated over all 333 locked rows, not only a tail preselected by the observed outcome. The score is negative predicted residual and the score threshold is fixed at zero before final evaluation. Table VI reports PR-AUC, ROC-AUC, balanced accuracy, precision, recall, and F1. These are diagnostics rather than Gate A components: they describe

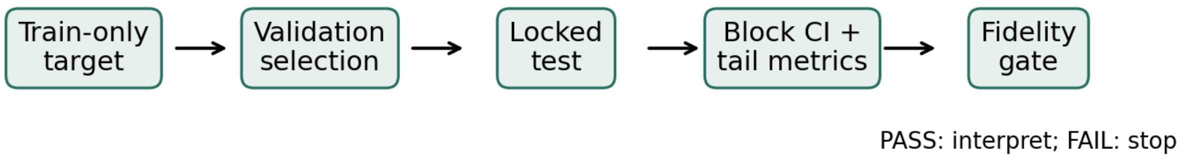

Fig. 1. Prospective audit workflow. Selection is complete before the locked 2016–2025 test; the final fidelity gate controls whether interpretation may proceed.

TABLE II
FIXED GATE POLICY. GATE A AND PRIMARY GATE B1 ARE INDEPENDENT; GATE B2 IS SENSITIVITY ONLY. MACHINE-READABLE VALUES ARE IN CONFIGS/FIDELITY_GATE.YAML.

Component	Role	Pre-specified pass condition
Gate A overall	Primary	Selected model vs. zero: upper paired 95% RMSE CI < 0
Gate A primary tail	Primary, $z < -1$	RMSE/MAE CI upper < 0; rank CI lower > 0 and permutation $p < .05$; top-10 lift > 1, lift-CI lower > 1, H and within-year P $p < .05$
Gate B1 weather value	Primary	Selected Weather only vs. Metadata only: upper paired 95% RMSE CI < 0
Gate B2	Sensitivity	Full vs. Metadata only; cannot replace Gate B1 or re-select
Severe tails	Sensitivity	$z < -1.5$ and $z < -2$; cannot replace the primary tail

TABLE III
VALIDATION SELECTION ON 2012–2015. RMSE IS COMPUTED AFTER FIXED-SEED PREDICTION AGGREGATION; SEED SD IS THE ACROSS-SEED RMSE STANDARD DEVIATION.

Model	Features	n	Seeds	RMSE	Seed SD	Selected
ExtraTrees (leaf=1)	Weather only	140	5	0.384	0.003	Yes
ExtraTrees (leaf=2)	Weather only	140	5	0.385	0.001	No
Random Forest (leaf=1)	Weather only	140	5	0.387	0.003	No
Random Forest (leaf=2)	Weather only	140	5	0.387	0.002	No
ExtraTrees (leaf=2)	Full	140	5	0.389	0.003	No
Random Forest (leaf=1)	Full	140	5	0.390	0.002	No

TABLE IV
VALIDATION STABILITY OF THE LEADING CONFIGURATIONS. THE RANK PROBABILITY IS ACROSS FIXED STOCHASTIC SEEDS; 1-SE DENOTES INCLUSION UNDER THE PRE-SPECIFIED ONE-STANDARD-ERROR DIAGNOSTIC, NOT A POST-HOC REPLACEMENT.

Model	Features	Mean RMSE	SD	P(rank=1)	1-SE
ExtraTrees (leaf=1)	Weather only	0.384	0.003	0.60	Yes
ExtraTrees (leaf=2)	Weather only	0.385	0.001	0.20	No
Random Forest (leaf=1)	Weather only	0.387	0.003	0.20	No
Random Forest (leaf=2)	Weather only	0.387	0.002	0.00	No
ExtraTrees (leaf=2)	Full	0.389	0.003	0.00	No
Random Forest (leaf=1)	Full	0.390	0.002	0.00	No

discrimination, but they do not override the paired-error and null-aware rank failures. In particular, an always-negative predictor cannot gain a gate pass through the negative-only sign rate used in older drafts.

C. Below-Trend Gate and Robustness

The broad below-trend subset contains 73 locked-test rows. Table VII separates error, rank-null, and chance-adjusted event-recovery components. The primary $z < -1$ top-10 lift is below random expectation and its permutation result is not significant; rank uncertainty also crosses zero. Gate A

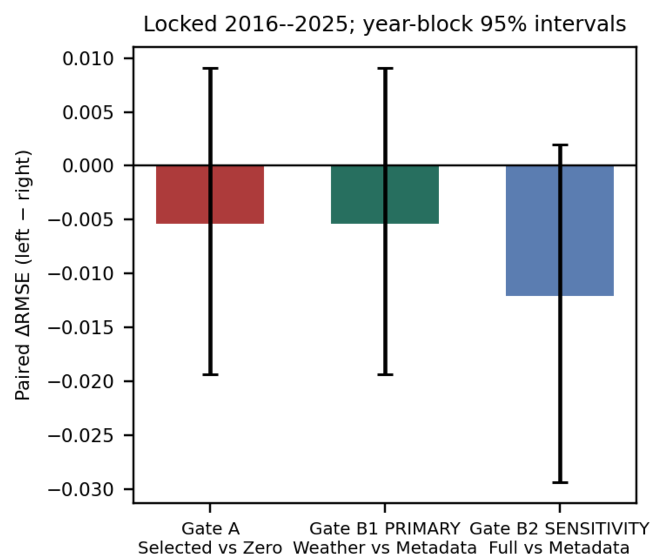

Fig. 2. Three distinct locked-test comparisons. Gate A evaluates predictive fidelity of the validation-selected weather-only model against the zero-residual baseline. Gate B1 is the primary incremental-weather comparison of the selected weather-only model against metadata-only context. Gate B2 compares the full and metadata-only models as a sensitivity analysis and cannot alter model selection. Values are paired ΔRMSE estimates with 95% calendar-year-block intervals; negative values favor the model on the left.

therefore fails without relying on a directional-sign statistic. Neither local explanations nor global/group diagnostics are evidence of weather causes or event drivers in this primary analysis.

Table IX gives exact temporal pass counts and summary ranges for the pre-specified stage-feature, crop, spatial, and

TABLE V

LOCKED 2016–2025 SAME-TASK AUDIT. LEARNED-MODEL PREDICTIONS ARE SEED AGGREGATED. ΔRMSE IS MODEL MINUS ZERO-RESIDUAL RMSE WITH PAIRED 95% YEAR-BLOCK INTERVALS; RMSE IS T HA^{-1}.

Model	Features	n	Seeds	R^2	RMSE	ΔRMSE vs zero [95% CI]
Crop train mean	Baseline	333	1	-0.031	0.674	0.000 [0.000, 0.000]
Train mean	Baseline	333	1	-0.031	0.674	0.000 [0.000, 0.000]
Zero residual	Baseline	333	1	-0.031	0.674	0.000
ExtraTrees	Full	333	5	0.006	0.662	-0.012 [-0.029, 0.002]
ExtraTrees	Metadata only	333	5	-0.031	0.674	0.000 [0.000, 0.000]
ExtraTrees	Weather only	333	5	-0.014	0.669	-0.005 [-0.019, 0.009]

TABLE VI

LOCKED-TEST EVENT-DETECTION DIAGNOSTICS ON ALL ROWS. THESE VALUES ARE NOT TUNED ON THE FINAL TEST AND DO NOT REPLACE GATE A COMPONENTS.

Tail	Prev.	PR-AUC	ROC-AUC	Bal. acc.	Precision	Recall	F1
z<-1	0.219	0.338	0.641	0.589	0.294	0.548	0.383
z<-1.5	0.156	0.294	0.672	0.611	0.228	0.596	0.330
z<-2	0.108	0.217	0.672	0.598	0.154	0.583	0.244

TABLE VII

TAIL COMPONENTS ON TRAIN-ONLY TARGETS. PANEL A REPORTS ERROR AND RANK RECOVERY; PANEL B REPORTS CHANCE-ADJUSTED TOP-10 RECOVERY. H IS THE HYPERGEOMETRIC PROBABILITY AND P IS THE WITHIN-YEAR PERMUTATION PROBABILITY. THE TOP-10 COMPONENT PASSES ONLY WHEN LIFT > 1, THE LOWER LIFT CI > 1, AND BOTH H AND P ARE $< .05$. ONLY $z < -1$ IS PRIMARY; SEVERE TAILS ARE SENSITIVITIES.

Panel A: Error and rank recovery

Role	Tail	n	ΔRMSE [95% CI]	ΔMAE [95% CI]	Rank ρ [95% CI]	Perm. p	Status
Primary	z<-1	73	-0.043 [-0.074, 0.007]	-0.054 [-0.096, 0.002]	0.180 [-0.035, 0.373]	0.362	FAIL
Sensitivity	z<-1.5	52	-0.056 [-0.083, -0.002]	-0.073 [-0.109, -0.007]	0.060 [-0.151, 0.416]	0.511	FAIL
Sensitivity	z<-2	36	-0.056 [-0.090, 0.014]	-0.076 [-0.118, 0.015]	-0.051 [-0.306, 0.497]	0.797	FAIL

Panel B: Chance-adjusted Top-10 recovery

Role	Tail	n	Observed	Expected	Lift [95% CI]	H p	P p	Status
Primary	z<-1	73	1/10	1.37	0.73 [0.00, 2.82]	0.794	0.793	FAIL
Sensitivity	z<-1.5	52	1/10	1.92	0.52 [0.37, 2.19]	0.907	0.935	FAIL
Sensitivity	z<-2	36	1/10	2.78	0.36 [0.32, 1.68]	0.979	0.985	FAIL

target-protocol checks. These checks do not reverse the central decision; they show its stability boundary rather than optimize around it.

D. Target, Cluster, and Model-Family Sensitivity

The three-observation rule is the pre-specified technical minimum, not an empirically optimal or stable variance threshold. Eligibility is audited at each validation/final cutoff. Histories 8 and 10 retain 313 rather than 333 final rows, so their favorable intervals are sensitivity results from a different population and cannot alter primary Gate A. Their eligible row IDs and target vectors are identical; their predictions differ only at floating-point precision (maximum absolute difference 1.11×10^{-16}), so the displayed metrics coincide. Table X separates target/model checks from resampling, so row counts and cluster counts cannot be confused.

The standardized-target model is actually refit on the train-only residual z, rather than obtained by rescaling raw-target predictions after the fact; its paired interval also crosses zero. Test-period z values need not have unit variance: each crop–state series uses its training-only residual scale, and a small training scale can yield larger future standardized residuals without leakage. Replacing ordinary least-squares detrending by a training-only Huber trend changes the primary event count from 73 to 40, but its paired interval still crosses zero. These analyses are deliberately reported as target-construction sensitivities, not candidate routes to a favorable result.

TABLE VIII
GATE DECISION COMPONENTS. GATE A CONTROLS SUBSTANTIVE OBSERVED-EVENT CLAIMS; PRIMARY GATE B1 IS ADDITIONALLY REQUIRED FOR WEATHER-SPECIFIC CLAIMS; GATE B2 IS SENSITIVITY ONLY.

Gate	Component	Scope	Status
Gate A	overall paired RMSE	locked final test	FAIL
Gate A	primary-tail paired RMSE	z<-1	FAIL
Gate A	primary-tail paired MAE	z<-1	FAIL
Gate A	rank recovery	z<-1	FAIL
Gate A	chance-adjusted top-k recovery	z<-1	FAIL
Gate B1	Selected Weather only vs. Metadata only	locked final test	FAIL
Gate B2	Full vs. Metadata only	locked final test	FAIL
Gate A	FINAL GATE A	z<-1	FAIL
Gate B1	FINAL GATE B1	weather-specific claim	FAIL

TABLE IX
ROBUSTNESS AND PROTOCOL SENSITIVITY. EXACT CRITERIA, ROW-LEVEL OUTPUTS, AND SETTINGS ARE RETAINED IN THE AUDIT ARTIFACT.

Audit	Main result	Role	Interpretation
Rolling origin	3/4 negative; 1/4 CI pass	Diagnostic	Unstable
Stage proxies	ΔRMSE -0.001	Sensitivity	No material change
Crop-specific	R^2 -0.273 to 0.072	Diagnostic	Unstable
Leave-one-state-out	4/12 states improve	Diagnostic	Unstable
Full-series detrending	Jaccard 0.605	Retrospective	Future yields enter target

TABLE X
TARGET AND MODEL SENSITIVITY. HISTORY 3 IS THE PRIMARY TECHNICAL MINIMUM. RMSE / ZERO GIVES MODEL AND ZERO-BASELINE RMSE. *SENSITIVITY PASS IS EVALUATED ON A DIFFERENT ELIGIBLE POPULATION AND CANNOT ALTER THE PRIMARY GATE A DECISION. HISTORY 8 AND HISTORY 10 HAVE IDENTICAL ELIGIBLE ROWS/TARGETS; THEIR MAXIMUM PREDICTION DIFFERENCE IS 1.11×10^{-16}, SO DISPLAYED METRICS COINCIDE.

Analysis	Role	Population	Rows	Scale	RMSE / zero	Delta RMSE [95% CI]	Result
History 3	Primary	Same	333	t ha^{-1}	0.669 / 0.674	-0.005 [-0.019, 0.009]	FAIL
History 5	Sensitivity	Same	333	t ha^{-1}	0.669 / 0.674	-0.005 [-0.019, 0.009]	FAIL
History 8	Sensitivity	Different	313	t ha^{-1}	0.530 / 0.553	-0.023 [-0.041, -0.007]	PASS*
History 10	Sensitivity	Different	313	t ha^{-1}	0.530 / 0.553	-0.023 [-0.041, -0.007]	PASS*
Standardized target	Sensitivity	Same	333	z units	3.530 / 3.500	0.030 [-0.009, 0.067]	FAIL
Huber detrending	Sensitivity	Same	333	t ha^{-1}	0.648 / 0.655	-0.007 [-0.022, 0.008]	FAIL
HistGradientBoosting	Sensitivity	Same	333	t ha^{-1}	0.675 / 0.674	0.001 [-0.016, 0.019]	FAIL
ElasticNet	Sensitivity	Same	333	t ha^{-1}	0.674 / 0.674	0.000 [-0.001, 0.001]	FAIL

E. Temporal Stability and Capacity

The four rolling-origin folds use non-overlapping future blocks after 2003, 2006, 2009, and 2012 training cutoffs. Three of four folds have negative point estimates; only one of four has a 95% CI fully below zero. A prefix learning curve retrains the same validation-locked configuration using progressively longer historical prefixes, while retaining only crop–state series with at least three prefix observations. None of the four prefix intervals establishes improvement over zero. This makes the conclusion conditional on the available panel and model family, not a claim that more data cannot help.

F. Audit Traceability and Interpretation Limits

Table XIII maps each numerical claim family to a concrete artifact. This is intentionally more than a code-release statement. E1 records the temporal split and train-only target construction; E2 records the entire validation grid, fixed seed outcomes, and tie rule; E3 stores every sampled calendar-year block; and E4 records every tail prediction, metric, and paired interval. The remaining checks preserve the temporal, feature, crop, spatial, target-construction, and baseline evidence required to reproduce the stop decision. Consequently, a reader can distinguish an empirical model result from a narrative inference about weather.

Several limits should shape interpretation. State-

TABLE XI

PAIRED RESAMPLING SENSITIVITY ON THE LOCKED 333-ROW POPULATION. THE NUMBER OF CLUSTERS IS A RESAMPLING PROPERTY, NEVER A ROW COUNT.

Resampling unit	Number clusters	Test rows	Delta RMSE	95% CI	Status
year block	10	333	-0.005	[-0.019, 0.009]	SENSITIVITY FAIL
state year block	120	333	-0.005	[-0.022, 0.010]	SENSITIVITY FAIL
crop state cluster	36	333	-0.005	[-0.033, 0.019]	SENSITIVITY FAIL

TABLE XII

EXACT TEMPORAL ROBUSTNESS AND PREFIX LEARNING-CURVE CHECKS. EACH ENTRY COMPARES THE LOCKED WEATHER-ONLY CONFIGURATION TO ZERO RESIDUAL WITH PAIRED CALENDAR-YEAR-BLOCK 95% INTERVALS.

Audit	n_{train}	n_{test}	ΔRMSE	95% CI	Status
Rolling fold 1 (2004–2006)	473	103	-0.010	[-0.039, 0.030]	FAIL
Rolling fold 2 (2007–2009)	561	102	0.017	[0.003, 0.028]	FAIL
Rolling fold 3 (2010–2012)	663	102	-0.015	[-0.090, 0.017]	FAIL
Rolling fold 4 (2013–2015)	769	105	-0.035	[-0.053, -0.025]	PASS
Prefix 25% through 1996	223	293	0.030	[0.020, 0.040]	FAIL
Prefix 50% through 2002	425	313	-0.011	[-0.028, 0.007]	FAIL
Prefix 75% through 2009	663	313	-0.016	[-0.036, 0.001]	FAIL
Prefix 100% through 2015	879	333	-0.005	[-0.019, 0.009]	FAIL

TABLE XIII

E1–E10 NUMERICAL TRACEABILITY. VERIFIED MEANS THE EVIDENCE ARTIFACT WAS REGENERATED AND CHECKED; IT DOES NOT MEAN A SCIENTIFIC GATE PASSED.

ID	Numerical trace	Status
E1	train-only target and split manifest	VERIFIED
E2	validation seed metrics and tie rule	VERIFIED
E3	2,000 paired year-block draws	VERIFIED
E4	all fixed tail metrics and intervals	VERIFIED
E5	four rolling-origin folds	VERIFIED
E6	stage-feature sensitivity	VERIFIED
E7	crop and leave-one-state-out checks	VERIFIED
E8	retrospective detrending sensitivity	VERIFIED
E9	baseline vectors, hashes, differences	VERIFIED
E10	numbers generated from audit records	VERIFIED

representative weather coordinates can miss within-state exposure; the weather covariates are correlated; and this panel lacks soil, irrigation, cultivar, management, and phenological measurements. The response variable is a detrended state aggregate rather than a farm-level loss. The observed gate failure may therefore reflect a mismatch among target, resolution, features, and model family, not an absence of physical weather effects. Conversely, these limitations do not justify relaxing the gate after observing the locked test. They identify inputs for a future, separately specified study: richer spatial covariates, prospectively defined phenology windows, and an independent evaluation period.

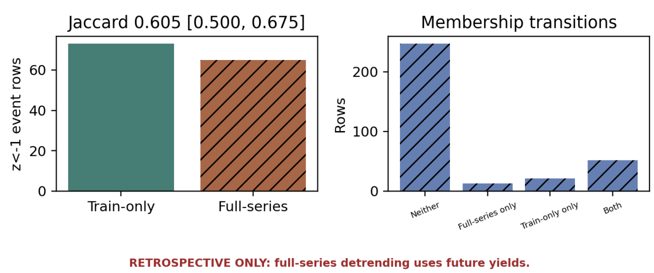

Fig. 3. **RETROSPECTIVE ONLY.** Full-series detrending uses future yields. Bars show exact event counts. Transition labels mean Neither, Train-only only, Full-series only, and Both; this is not prospective forecast evidence.

G. *Retrospective Target Sensitivity*

Full-series detrending uses later yields to construct earlier residuals. It is therefore a retrospective diagnostic, not a prospective target. Relative to the train-only construction, the $z < -1$ final-period set has Jaccard overlap 0.605 with year-block 95% interval [0.500, 0.675]; 21 prospective events are not retrospective events and 13 retrospective events are not prospective events (Fig. 3). A coherent retrospective explanation would still not repair the prospective fidelity failure.

V. DISCUSSION

A. *Claim Scope and Multiplicity*

The audit distinguishes primary decisions from descriptive measurements before examining the locked period. Gate A is a conjunction on the selected weather-only model, overall paired error, and the $z < -1$ tail; primary Gate B1 is the independent incremental-weather-value comparison that cannot alter selection. Gate B2 is sensitivity only and also cannot alter selection. The two more severe tails, alternative minimum-history rules, Huber detrending, standardized targets, rolling folds, prefix learning curves, feature-stage proxies, crop/state checks, and additional model families are sensitivity or diagnostic analyses. They provide uncertainty about the protocol boundary, but they are not multiple opportunities to declare the primary claim successful. No sensitivity replaces the validation-locked configuration, primary tail, or test population. This scope statement is important because an isolated favorable interval, such as a later-history or one rolling-origin result, is expected among many correlated analyses and does not repair the pre-specified gate failure.

B. Reproducibility and Traceability

The audit records the source-input hashes, target construction, locked configuration, fixed seeds, row identifiers, predictions, paired bootstrap draws, and table inputs. Each numerical claim is traceable to an evidence file; VERIFIED means regenerated and checked, never scientifically favorable. An anonymized artifact is prepared for release where permitted by the venue and source licenses.

Traceability is especially important here because several quantities that can look interchangeable are not. The raw residual in t ha^{-1} is the prediction target; the training-only standardized residual defines the event threshold; and a cluster count is the unit sampled for uncertainty, not the number of test rows. The released records retain the row-aligned target and prediction vectors, zero-residual baseline, calendar-year bootstrap draws, and configuration hash. A reader can therefore recompute a reported difference in RMSE, inspect its confidence interval, and check that both sides of a comparison use identical locked rows. This design also makes a failed check informative: an artifact can be complete and a gate can still fail.

The selection boundary is similarly explicit. Candidate models, feature families, and seeds are evaluated only in validation, after which one weather-only ExtraTrees configuration is locked. The final period is then used for the paired Gate A/B audit rather than for a second search. Alternative target constructions, histories, resampling units, and model families are preserved with their roles in the evidence table. A favorable sensitivity is consequently visible but cannot be silently promoted into the primary conclusion. This is a practical guard against multiple correlated analyses creating an attractive but unsupported narrative.

The event metrics follow the same boundary. Precision, recall, F1, balanced accuracy, PR-AUC, and ROC-AUC are calculated over every locked row using the fixed zero score threshold, with the confusion-matrix counts retained in the artifact. They describe discrimination under a fixed operational rule, whereas Gate A asks a stricter question: whether the selected model improves paired error and recovers the primary tail beyond the specified null-aware checks. No collection of diagnostics substitutes for that conjunction. This distinction is why the paper reports a descriptive event table while withholding event-driver interpretation.

C. Limitations and External Validity

State-representative weather can miss within-state exposure, weather covariates are correlated, and the panel lacks soil, irrigation, cultivar, management, and phenology data. This negative result limits the present target, resolution, features, and model family; it does not establish that weather has no yield effect. A future study needs prospectively versioned inputs, finer exposure measurements, and an independent future test interval.

We also implemented a separate county-level winter-wheat check rather than assuming that a finer panel would rescue the state-level result. The locked pre-model population contains 574 counties with at least ten training-period observations from 2000–2025, official county yields, static county interior points, and annual NASA POWER weather aggregates. Validation selected a weather-only ExtraTrees residual model. On the untouched 2022–2025 holdout, its RMSE was 13.47 versus 13.78 bushels per acre for the zero-residual baseline, but the paired year-block 95% CI for the difference was $[-0.67, 0.26]$. Gate A therefore remains inconclusive, and the county workflow emits ABSTAIN. This is an additional reproducible agricultural abstention case, not a successful agricultural prediction result or evidence of transfer.

The temporal and population sensitivities also limit external validity. Three of four rolling folds have negative point estimates, but only one has a confidence interval entirely below zero; the direction and uncertainty are therefore not stable enough for a general predictive claim. Raising the minimum history from three to eight or ten years changes eligibility from 333 to 313 locked rows. Its favorable interval is useful evidence that series history matters, but it estimates a different population and is labeled as such. Full-series detrending changes target membership by using future yields and remains retrospective only. These boundaries identify what should be pre-specified and externally validated before a later study revisits the predictive or explanatory question.

VI. Synthetic and External-Domain Evidence

The stop-gate policy is also evaluated where ground truth is available. A deterministic synthetic temporal benchmark covers no, weak, moderate, and strong signal; correlated features; omitted confounding; temporal and geographic shift; leakage; small samples; imbalanced tails; measurement error; and spatial-resolution mismatch. Ungated explanations permit attribution in every scenario, including invalid regimes. The full gate reduces simulated false permission from 28.6% to 0% while permitting explanation in six valid-signal scenarios. Thus the gate is not an always-abstain rule; it is a conditional permission rule.

An external, non-agricultural stress test uses public daily PJM electricity-demand records. Calendar-only features are fit on January–September 2024 and evaluated on a locked October–December period. Adding the day-ahead demand forecast improves locked RMSE from 227,417 to 77,521 MWh, with paired bootstrap ΔRMSE 95% CI $[-186,037, -119,309]$. The corresponding feature-group gate passes, so permutation importance is made available for this external forecasting task. This result establishes that the procedure can permit interpretation when its pre-specified predictive and incremental-value conditions are met. It is neither a causal claim nor evidence of agricultural transfer.

VII. Conclusion

This paper evaluates whether an interpretable weather-only model can support substantive claims about below-trend U.S. crop-yield events. The model and feature family were selected before a locked 2016–2025 test. Gate A requires paired predictive fidelity and null-aware recovery of the primary $z < -1$ tail; primary Gate B1 requires incremental weather value of the selected weather-only model over metadata. Gate A and Gate B1 fail. A separately locked county-level winter-wheat panel also remains inconclusive under Gate A, despite a favorable point estimate. Accordingly, the permitted conclusion is procedural and negative: this fitted model may be described, but its explanations are not evidence of observed event drivers or weather-specific effects in this panel. The reusable lesson is to make data availability and target construction explicit, lock selection before testing, compare same-row baselines with paired uncertainty, distinguish predictive fidelity from feature value, and stop interpretation when a pre-specified gate fails. The audit artifacts preserve the numerical path to that decision.

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